

HAT-P-36b

R-1733

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_200201 · created 2026-08-02T17:49:20+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458880.8883 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1143 +/- 0.0095
Transit depth (Rp/Rs)^2	1.31 +/- 0.22 [%]
Orbital Inclination (inc)	85.29 +/- 2.24
Airmass coefficient 1 (a1)	1.116 +/- 0.011
Airmass coefficient 2 (a2)	-0.1 +/- 0.01
Scatter in the residuals of the lightcurve fit is	0.99 %
Best Comparison Star	None
Optimal Aperture	4.14
Optimal Annulus	19.65
Transit Duration (day)	0.0904 +/- 0.0058

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1143 $\pm$ 0.0095 vs 0.126 (deviation 1.2 σ)
Duration fit vs published	0.0904 d vs 0.0925 d (deviation 0.35 σ)
Mid-time O-C	-8.5 min
Depth SNR	11.7
Residual scatter	0.99 %

Light curve

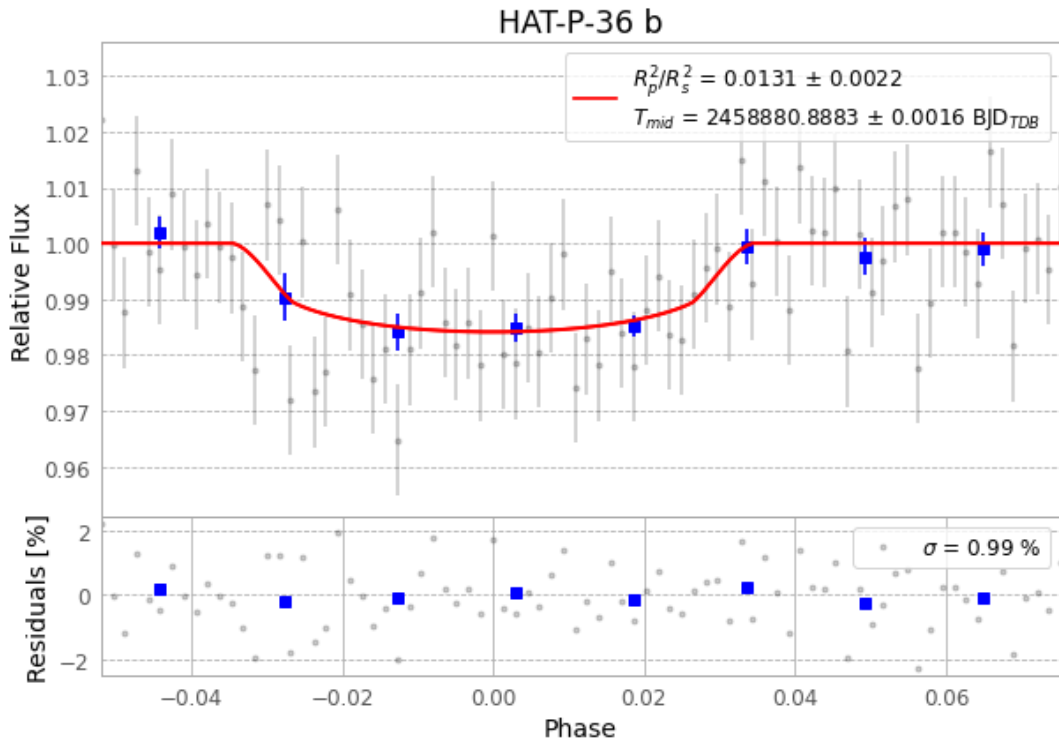


Figure 1 — FinalLightCurve_HAT-P-36 b_2020-02-01.png (SHA-256 da163255266730c3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [445, 207], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ef3712d6a42a774c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

82 FITS frames (54 MB), HATP-36200201073312.FITS → HATP-36200201113613.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-200201.log (957ddbc16b3d...), FinalLightCurve_HAT-P-36 b_2020-02-01.png (da1632552667...), FinalLightCurve_HAT-P-36 b_2020-02-01.csv (a1cc901416e0...)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)